

MONSTER PACKS

Game Design Document

com.sergivm.monsterpacks

Version 0.4 — UX/UI Design Integrated

Single source of truth. Includes all design, values, and technical decisions.

1. Technical Stack

Android-only, Kotlin + Jetpack Compose. No Java, no LiveData, no GlobalScope. All async via Coroutines + Flow/StateFlow.

Technology	Version	Notes
Kotlin	1.9.22	Primary language. No Java.
AGP	8.3.0	Do not update without reviewing KSP compatibility.
KSP	1.9.22-1.0.17	Must match Kotlin version exactly.
minSdk	28 (Android 9)	Minimum supported.
targetSdk / compileSdk	34	Current target.
Jetpack Compose BOM	2024.02.00	Manages Compose versions internally.
Hilt	2.50	Dependency injection.
Room	2.6.1	Local persistence.
Coroutines	1.7.3	Game loop and async operations.
Gson	2.10.1	GameState serialisation to JSON for Room.
DataStore	1.0.0	Lightweight config and preferences.
Navigation Compose	2.7.7	Screen navigation.

1.1 Architecture — Clean Architecture (Strict)

- domain/model/ — All entities and models. Each in its own file.
- domain/engine/ — GameEngine: pure logic, zero Android or Compose imports.
- domain/usecase/ — Optional use cases for complex logic.
- data/db/ — Room entities, DAOs, Database.
- data/repository/ — GameRepository: load and save game state.
- presentation/viewmodel/ — Hilt ViewModels connecting engine to UI.
- presentation/screen/ — Compose screens.
- presentation/ui/theme/ — Theme, colours, typography.
- app/di/ — Hilt modules.

⚠ GameEngine must NOT import anything from Android or Compose. Pure logic, fully testable with JUnit.

⚠ No business logic in ViewModels. No logic in Compose. Never use LiveData — Flow and StateFlow throughout. Never block main thread with Room — always Dispatchers.IO.

2. Game Overview

Monster Packs is an offline mobile card-collecting game. Players open packs to discover creatures, real animals, plants, objects, landscapes, and legendary beings — completing collections and levelling up. No internet connection required. Visual style: colorful, epic, modern, detailed, non-minimalist.

2.1 Core Loop

- Open the free Basic Pack — unlimited, no cooldown.
- Cards revealed one by one in ascending rarity order (least rare first, most rare last).
- Earn Coins and Gems from each card obtained.
- Earn XP per pack opened — level up to unlock upgrades.
- Spend resources on Special Packs or upgrades in the Shop.
- Complete collections.

2.2 First Launch — Username Setup

On first launch, before accessing any screen, the player must enter a username. This name is displayed in the top bar across all main screens. The username can be changed once from Settings > Account. No further changes are permitted.

2.3 Navigation — Bottom Tab Bar

Tab	Icon	Description
Pack		Main free pack opening screen.
Collection		View all cards, unlocked and locked.
Shop		Upgrades and special packs.
Settings		Language, Google Play, account, legal.

i The bottom navigation bar is hidden during pack opening (from the moment the player taps 'Open a Pack' until the summary screen is dismissed).

3. Models

Cards are pre-defined static entities. Opening a pack reads from the existing pool and updates the player's copy count. Pack sessions are not persisted — only card counts and resources matter.

3.1 Card

Field	Type	Description
id	Int	Unique identifier.
name	String	Display name.
type	CardType (enum)	Thematic family. Determines background colour.
rarity	Rarity (enum)	Drop probability, shape, animation, rewards.
collection	Collection (ref)	Card pool this card belongs to.
description	String?	Short flavour text (1–2 sentences). Present for Common, Rare, Epic. Null for Special, Legendary, God.
imageRes	String / ResId	Placeholder or final illustration.
collectionNumber	Int	Card number within the collection (e.g. #042). Shown on card bottom-left.
coinReward	Int	Coins awarded on obtain.
gemReward	Int	Gems awarded (0 for Common / Rare / Epic).
copies	Int	Player-owned copies. 0 = locked (silhouette shown).
isVariant	Boolean	True if this is a Special / Legendary / God variant of a base card.
baseCardId	Int?	Reference to the base card if isVariant = true.

i Card descriptions are unique per card — short flavour text that describes the creature in 1–2 sentences. Shown during opening and in the collection detail view. Not shown on Special, Legendary, or God cards (their art speaks for itself).

3.2 PackDefinition

Field	Type	Description
id	String	Unique identifier (e.g. 'pack_basic', 'pack_fire').
name	String	Display name.

type	PackType (enum)	BASIC, TYPE_THEMED, RARITY_BOOSTED, LUCKY, REWARD.
collection	Collection (ref)	Card pool source.
cardPool	List<CardType>?	TYPE_THEMED only: restricts to these types.
slotRules	List<SlotRule>	Per-slot rarity rules or roll distribution.
cost	PackCost?	Null for free packs; otherwise Coins or Gems.
coverRes	String / ResId	Collection cover art shown on pack front.
designRes	String / ResId	Pack border/form asset — unique per pack type.
backgroundRes	String / ResId	Background shown during opening — set by collection and pack type.
xpReward	Int	XP granted when this pack is opened.

i Each pack has: (1) a collection cover for the front face, (2) borders and decorative forms that vary by pack type, (3) a background that changes during opening based on collection and type.

i A pack can never contain two copies of the same card. The roll engine must re-roll if a duplicate would appear within the same opening session.

3.3 SlotRule

Field	Type	Description
slot	Int	Slot index (1–5 for standard packs, 1–3 for Bonus Pack).
guaranteedRarity	Rarity?	If set, this slot always yields exactly this rarity.
minimumRarity	Rarity?	If set, roll cannot produce below this rarity (Lucky Packs).
rollTable	Map<Rarity, Float>?	Custom probability table for free-roll slots.

3.4 Collection

Field	Type	Description
id	String	Unique identifier.
name	String	Display name.
cards	List<Card>	All cards in this collection (base + variants).

coverRes	String / ResId	Cover image shown on pack face and collection header.
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3.5 PlayerState

Field	Type	Description
username	String	Chosen on first launch. Changeable once from Settings > Account.
usernameChanged	Boolean	True if the username has already been changed once. Blocks further changes.
coins	Long	Current coin balance.
gems	Long	Current gem balance.
xp	Long	Total accumulated XP.
level	Int	Current player level.
cardCopies	Map<Int, Int>	cardId -> copy count.
rewardPackReadyAt	Long?	Epoch ms when next Bonus Pack is ready.
basicPackUpgradeLevel	Int	Upgrade tier for Basic Pack.
rewardPackUpgradeLevel	Int	Upgrade tier for Bonus Pack.

4. Card Types

10 types in the initial collection. Card counts are intentionally unequal — some types feel more exclusive. Types can contain fantastical creatures, real animals, plants, objects, and real-world phenomena. Future versions may introduce fully real-world types (e.g. 'Dogs').

Type	Theme	Colour (hex)	Card Count
Infernal	Fire, volcanoes, magma, hell	#C0392B (deep red)	15
Abyssal	Deep sea, darkness, pressure	#1A3A5C (dark navy)	14
Spectral	Death, dark magic, undead	#4A235A (dark purple)	14
Celestial	Stars, cosmos, divine light	#D4AC0D (gold)	15
Storm	Lightning, wind, tempests	#2874A6 (electric blue)	14
Verdant	Forest, nature, plants, earth	#1E8449 (forest green)	13
Frost	Ice, tundra, arctic creatures	#AED6F1 (ice blue)	14
Arcane	Pure magic, runes, constructs	#7D3C98 (violet)	14
Savage	Wild beasts, primal instinct	#D35400 (burnt orange)	14
Mythic	Gods, ancient myths, legends	#B7950B (antique gold)	15

5. Rarities

Rarity controls drop probability, card visual shape/frame, animation tier, rewards, and icon. Cards are always revealed in ascending rarity order within a pack — least rare first, most rare last.

Rarity	Icon	Coin Reward	Gem Reward	Base Drop % (free slots)	Animation Tier
Common	✱	1	0	~58%	Simple fade-in
Rare	⚡	2	0	~35%	Glow pulse
Epic	💥	10	0	5%	Particle burst
Special	★	100	10	1%	Full-screen shatter
Legendary	💧	500	50	0.1%	Epic cinematic
God	🪐	2000	200	0.01%	Full legendary sequence

Rarity icons are displayed on the card face (bottom-right area alongside the card number) and in the Collection screen progress counters.

5.1 Variant Card Rules

Variant Tier	Based on	Visual Treatment	Rule
Special	Rare or Epic	Enhanced illustration, special background effects	Selected Rares and Epics get a Special variant
Legendary	Epic only	Full-art — card border dissolves into the artwork	Only selected Epics get a Legendary variant
God	Common, Rare, or Epic	Unique layout, representative of entire type	Exactly one God per type (10 total)

i Common cards never have Special or Legendary variants. Descriptions (flavour text) are shown only on Common, Rare, and Epic cards. Special, Legendary, and God cards have no description — their artwork is the statement.

6. Card Catalogue — Initial Collection

Complete proposed card list. The Description column contains the flavour text shown on the card during opening and in the collection detail. Special, Legendary, and God cards have no description field — the notes column describes the art direction instead.

6.1 Infernal (15 cards)

Card Name	Rarity	Description / Art Direction
Ember Lizard	Common	Small fire lizard, base creature. Skitters across volcanic rock leaving scorch marks.
Ash Crow	Common	Charred black crow with ember eyes. Feeds on the remains of those consumed by flame.
Cinder Beetle	Common	Volcanic beetle with a glowing carapace. Its shell is hot enough to brand skin.
Smolder Toad	Common	Volcanic toad that spits bursts of fire. Found near geothermal vents.
Flamehorn Ram	Common	Ram with permanently burning horns. Charges headfirst into anything that moves.
Pyro Hellhound	Rare	Classic hellhound with a mane of living fire. Its howl ignites the air around it.
Magma Crab	Rare	Crab armoured in cooled lava rock. Crushes prey with claws that reheat on impact.
Lava Serpent	Rare	A serpent that swims through magma as if it were water.
Lava Serpent (Special)	Special	Full-art redesign. Glowing lava veins pulse beneath obsidian scales.
Cinderclaw Wyvern	Epic	Wyvern with talons of molten rock. Leaves rivers of fire wherever it lands.
Cinderclaw Wyvern (Legendary)	Legendary	Full-art, obsidian and fire legendary frame. The sky behind it is permanently ash.
Molten Serpent	Epic	A massive serpent of pure magma. Older than the volcanoes it inhabits.
Molten Serpent (Special)	Special	Enhanced art with heat distortion effects rippling across the frame.
Ember Phoenix	Epic	A phoenix reborn endlessly from its own ash. Its tears are liquid fire.
Magma Titan (GOD)	God	God of Infernal. A colossal volcanic titan. Its footsteps open new craters.

6.2 Abyssal (14 cards)

Card Name	Rarity	Description / Art Direction
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Brine Shrimp	Common	A tiny deep-sea shrimp with bioluminescent spots. Harmless alone. Lethal in swarms.
Cave Eel	Common	A blind eel from flooded underwater caverns. Navigates by pressure alone.
Deepwater Crab	Common	A real deep-sea crab grown to impossible size under crushing pressure.
Fangtooth Fish	Common	A real deep-sea fish with teeth too large to close its mouth. Pure terror, small scale.
Pressure Jellyfish	Common	A translucent jellyfish that implodes violently if brought to the surface.
Abyssal Manta	Rare	An enormous manta ray gliding through total darkness. Its wingspan blocks the sonar.
Voidfin Shark	Rare	A shark adapted to lightless depths. Its skin absorbs all light around it.
Lurker Eel	Rare	A massive predatory eel that waits motionless for centuries before striking.
Bioluminescent Jellyfish	Rare	A stunning glowing jellyfish inspired by real deep-sea species. Beautiful and lethal.
Bioluminescent Jellyfish (Special)	Special	Full-art. Glowing tentacles fill the entire frame in electric blue and violet.
Deep Horror	Epic	A Lovecraftian deep-sea monstrosity. Its shape changes depending on who is watching.
Deep Horror (Legendary)	Legendary	Full-art legendary. Tentacles extend beyond the card border into the void.
Leviathan	Epic	The classic sea serpent of legend. Long enough to encircle a continent.
The Kraken (GOD)	God	God of Abyssal. The legendary kraken. Its tentacles have sunk entire fleets.

6.3 Spectral (14 cards)

Card Name	Rarity	Description / Art Direction
Shadow Bat	Common	A small bat made of living shadow. Disappears completely in darkness.
Grave Moth	Common	A moth that feeds on the moss growing on ancient tombstones. Ancient and quiet.
Pale Wisp	Common	A floating ghostly orb of cold light. Lures travelers away from safe paths.
Bone Rat	Common	A skeletal rat. Undead vermin that gnaws on things it can no longer taste.
Dusk Crow	Common	A crow that vanishes completely at daybreak. Only its call remains at dawn.
Specter Wolf	Rare	A translucent ghostly wolf. Its howl can only be heard by those about to die.

Grave Gargoyle	Rare	A stone gargoyle animated by dark magic. Returns to stone when it believes it is watched.
Phantom Cat	Rare	A cat that phases through walls. It has lived in the same house for three centuries.
Duskborn Vampire	Epic	A classic vampire, elegant and deadly. Has outlived every civilization it has fed from.
Duskborn Vampire (Special)	Special	Full-art, crimson moon backdrop. Fangs catch the light of a dying star.
The Wraith	Epic	An incorporeal dark entity that passes through all matter. It has no memory of what it was.
The Wraith (Legendary)	Legendary	Full-art. Shadow tendrils consume the entire frame. Only eyes remain.
Pale Banshee	Epic	A wailing spirit that foretells death. Its scream is heard three days before the event.
Lich of the Abyss (GOD)	God	God of Spectral. An undying lich robed in the souls of ten thousand fallen.

6.4 Celestial (15 cards)

Card Name	Rarity	Description / Art Direction
Stardust Moth	Common	A moth with wings patterned like a distant galaxy. Navigates by starlight alone.
Comet Fox	Common	A small fox trailing a permanent stream of stardust. Appears only during meteor showers.
Moon Hare	Common	A rabbit that only becomes visible under direct moonlight. Invisible at noon.
Astral Firefly	Common	A firefly glowing in nebula colours — violet, gold, and deep blue.
Cloud Turtle	Common	A turtle drifting slowly through the upper atmosphere. It never descends.
Nebula Serpent	Rare	A serpent with galaxy-patterned scales. Every scale reflects a different star system.
Astral Pegasus	Rare	A winged horse with a mane of star-trail light. Leaves constellations in its wake.
Meteor Golem	Rare	A golem built entirely from meteorite fragments. Hotter to the touch than it appears.
Lunar Moth	Rare	A giant moth drawn powerfully to lunar energy. Grows larger each full moon.
Lunar Moth (Special)	Special	Full-art. The moon glows through translucent wings mapped with light.
Stardust Unicorn	Epic	A unicorn with a horn of condensed starlight. Its blood is liquid aurora.
Stardust Unicorn (Special)	Special	Enhanced art. A galaxy is reflected perfectly in both of its eyes.

Solar Griffin	Epic	A griffin radiating solar energy. Blind to look at directly. Warm to stand near.
Solar Griffin (Legendary)	Legendary	Full-art. An eclipse halo surrounds the legendary frame. Light bends around it.
Dawn Phoenix (GOD)	God	God of Celestial. A phoenix of pure sunlight. Its rebirth creates new stars.

6.5 Storm (14 cards)

Card Name	Rarity	Description / Art Direction
Gale Sparrow	Common	A small bird that rides the leading edge of storm fronts. Never lands willingly.
Storm Cloud	Common	A living storm cloud — a real meteorological phenomenon given consciousness and hunger.
Thunder Gecko	Common	A lizard that stores static charge in its tail. Shocks anything that touches it.
Squall Bat	Common	A bat that navigates by lightning echolocation during active electrical storms.
Rain Serpent	Common	A serpent that appears only during heavy rainfall. Dissolves into water at storm's end.
Gale Hawk	Rare	A hawk that generates small tornados at its wingtips with every downstroke.
Lightning Eel	Rare	A massively oversized electric eel. Generates enough charge to stop a heart at distance.
Cyclone Drake	Rare	A small drake that spawns miniature cyclones when it exhales.
Cyclone Drake (Special)	Special	Full-art. The card shows the view from inside the eye of a storm.
Storm Giant	Epic	A colossal humanoid born from a supercell storm. Speaks only in thunder.
Storm Giant (Special)	Special	Enhanced art. Lightning veins trace paths across the giant's stone skin.
Thunderbird	Epic	From Native American myth. Creates thunder with every wingbeat. Cannot be calm.
Thunderbird (Legendary)	Legendary	Full-art legendary. A storm crown. Every feather is a living lightning bolt.
Tempest Lynx (GOD)	God	God of Storm. A lynx of living electricity. It is the storm. The storm is it.

6.6 Verdant (13 cards)

Card Name	Rarity	Description / Art Direction
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Moss Turtle	Common	A real turtle with a full ecosystem of moss growing on its shell. Very slow. Very old.
Bramble Rabbit	Common	A rabbit perfectly camouflaged inside thornbush. Entirely invisible when still.
Spore Mushroom	Common	A sentient mushroom that releases clouds of disorienting spores when threatened.
Petal Sprite	Common	A tiny fairy made entirely of living petals. Falls apart in autumn. Regrows in spring.
Vine Frog	Common	A frog that blends seamlessly into rainforest vines. Hunts by absolute stillness.
Thorn Wolf	Rare	A wolf with thorned fur that functions as natural armour. Impossible to grab.
Root Serpent	Rare	A serpent that burrows through root systems and emerges from the earth without warning.
Mushroom Golem	Rare	A golem of compressed mycelium. Rebuilds itself from any nearby fungi when damaged.
Mushroom Golem (Special)	Special	Full-art. A spore cloud fills the entire background in deep green and gold.
Ancient Treant	Epic	A thousand-year-old tree that has woken up. Its voice is the sound of roots splitting stone.
Ancient Treant (Legendary)	Legendary	Full-art. An entire old-growth forest exists inside the card frame.
Vine Hydra	Epic	A hydra of living vines. Each severed head regrows as two new branches by morning.
Bramble Bear (GOD)	God	God of Verdant. A bear woven from the entire forest. As old as the soil beneath it.

6.7 Frost (14 cards)

Card Name	Rarity	Description / Art Direction
Snowdrift Owl	Common	An owl with feathers of compressed snow. Silent and weightless. Impossible to track.
Ice Hare	Common	An arctic hare with a white coat and frost breath. Real species, mythologised.
Glacier Crab	Common	A crab encased in living ice armour that continuously regenerates if cracked.
Frost Viper	Common	A snake that freezes prey solid with a single bite. Hunts in blizzards.
Tundra Fox	Common	An arctic fox, real species, elevated to myth. Said to guide lost travelers to warmth.
Permafrost Yeti	Rare	A yeti adapted to the deepest polar ice. Has not been warm since the last ice age.
Arctic Kitsune	Rare	A nine-tailed fox of the frozen north. Each tail controls a different aspect of winter.

Blizzard Serpent	Rare	A serpent that travels inside blizzards. Indistinguishable from the storm itself.
Arctic Kitsune (Special)	Special	Full-art. Nine tails fan across a frozen aurora in silver and pale blue.
Ice Sphinx	Epic	A sphinx carved from a living glacier. Still. Patient. Unspeakably old. Still alive.
Ice Sphinx (Special)	Special	Enhanced art. Ice crystals refract light into a full spectrum across the frame.
Glacier Wurm	Epic	A dragon-like wurm of pure glacial ice. Its breath locks the sea solid in seconds.
Glacier Wurm (Legendary)	Legendary	Full-art legendary. An entire ice age is contained within a single card.
Frost Giant (GOD)	God	God of Frost. The primordial frost giant. The first winter began the moment it woke.

6.8 Arcane (14 cards)

Card Name	Rarity	Description / Art Direction
Mana Wisp	Common	A floating orb of raw magical energy. Harmless until it destabilises. Then catastrophic.
Hex Cat	Common	A cat that accidentally casts minor spells when it blinks. No malice. Constant chaos.
Runic Beetle	Common	A beetle with runes carved naturally into its shell. Each rune does something different.
Glyph Moth	Common	A moth whose wing patterns are live, shifting runes. Impossible to study. Always moving.
Scroll Serpent	Common	A serpent that wraps itself into scroll form. Libraries treat them as pests.
Glyph Drake	Rare	A small drake that breathes runic fire. Whatever it burns becomes marked with power.
Spellbound Gargoyle	Rare	A gargoyle animated by a spell that has since been forgotten. Cannot be dispelled.
Arcane Chimera	Rare	A three-headed magical construct chimera. Each head belongs to a different school of magic.
Arcane Chimera (Special)	Special	Full-art. Each head radiates a distinct magical school in colour and energy.
Ether Serpent	Epic	A serpent that exists partially in another dimension. You can see through it. It can see through you.
Ether Serpent (Special)	Special	Enhanced art. The serpent is mid-phase between planes, splitting the frame in two realities.
Runic Golem	Epic	A massive golem constructed from stacked arcane runes. Each rune is a binding law.
Runic Golem (Legendary)	Legendary	Full-art legendary. Every rune on its body is legible. Together they tell a complete story.

Construct Prime (GOD)	God	God of Arcane. The original magical construct. The source from which all spells were copied.
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6.9 Savage (14 cards)

Card Name	Rarity	Description / Art Direction
Dire Boar	Common	A massive wild boar at prehistoric scale. Nothing moves it. Nothing stops it.
Razorback Wolf	Common	A wolf with blade-like dorsal spines that emerge when it runs. Leaves a trail of cuts.
Ironhide Armadillo	Common	An armadillo with near-indestructible natural plating. Real species, mythologised.
Tusk Badger	Common	A badger with elongated ivory tusks. Ferocious beyond any reasonable size.
Primal Crow	Common	A crow from before recorded history. It remembers things that no longer exist.
Feral Lynx	Rare	A lynx that has learned to hunt supernatural prey. Patient. Precise. Relentless.
Saber-Tooth Tiger	Rare	The classic prehistoric apex predator. Still alive. Still hunting. Still hungry.
Dire Bear	Rare	A bear the size of a small building. Its growl registers as seismic activity.
Saber-Tooth Tiger (Special)	Special	Full-art. An ice age landscape stretches behind it under a pale, prehistoric sky.
Bone Crusher Ape	Epic	An ape of pure prehistoric muscle. Its knuckles have cracked stone floors to rubble.
Bone Crusher Ape (Special)	Special	Enhanced art. Knuckles mid-impact, stone fragmenting in every direction.
Primal Raptor	Epic	A raptor that survived beyond its own extinction through pure ferocity and spite.
Primal Raptor (Legendary)	Legendary	Full-art legendary. A hunting pack moves in formation behind it through primeval fern.
Ironhide Rhino (GOD)	God	God of Savage. The first beast. Older than memory. Older than names.

6.10 Mythic (15 cards)

Card Name	Rarity	Description / Art Direction
Harpy	Common	A winged woman of Greek myth. Steals food and sanity in equal measure.
Satyr	Common	Half-man, half-goat. Deeply mischievous. Deeply committed to revelry. Unavoidable.

Siren	Common	A dangerous sea singer. Her voice turns ships toward rocks. She never means harm.
Imp	Common	A small demonic trickster. Its pranks begin minor. They never end minor.
Centaur Scout	Common	A young centaur with a bow. First of its kind to venture beyond the forest's edge.
The Cyclops	Rare	A one-eyed giant of Greek legend. Slow to anger. Catastrophic when angry.
Medusa	Rare	A gorgon whose gaze petrifies. Her garden of statues is the most visited in the world.
Sphinx of Riddles	Rare	The Egyptian sphinx. Keeper of ancient questions. Deadly to those who answer wrong.
Medusa (Special)	Special	Full-art. Stone statues of failed challengers surround her in the background.
The Minotaur	Epic	The labyrinth beast. Half-man, half-bull. Still pacing. Still waiting. Still hungry.
The Minotaur (Special)	Special	Enhanced art. Maze corridors recede infinitely behind the beast.
Chimera	Epic	A lion-goat-serpent, three natures permanently at war inside a single body.
Chimera (Legendary)	Legendary	Full-art legendary. All three natures are visible, each pulling in a different direction.
Cerberus	Epic	Three-headed guardian of the underworld. Has never once let anyone leave.
Hydra (GOD)	God	God of Mythic. The immortal Hydra. Every head is a different legend. None can be silenced.

7. Pack System

All packs: no duplicate cards per session, cards revealed in ascending rarity order. Each pack has a unique visual identity: collection cover on the front face, type-specific border and decorative forms, and a collection+type background during opening.

7.1 Basic Pack (Free, Unlimited)

Slot	Rule
Slot 1	Always Common
Slot 2	Always Common
Slot 3	Always Rare (exactly)
Slot 4	Free roll: ~58% Common, ~35% Rare, 5% Epic, 1% Special, 0.1% Legendary, 0.01% God
Slot 5	Same free roll as Slot 4

7.2 Bonus Pack (Free, Cooldown-Based)

Generates automatically in the Shop. All parameters are upgradeable (requires sufficient player level).

Parameter	Default Value	Upgradeable?
Cards per pack	3	Yes
Gems per pack	5	Yes
XP per pack	5	Yes
Cooldown	10 minutes	Yes — reduced per upgrade tier
Max stored packs	1	Yes — increased per upgrade tier
Card rarity table	Standard free roll	Yes — improved per upgrade tier

7.3 Special Packs (Paid)

7.3.1 Type-Themed Packs (10 total — one per type)

Restricts the card pool to that type. Standard rarity distribution. Players targeting a specific type use these.

7.3.2 Rarity-Boosted Packs

Shifts the probability table toward a target rarity — does not guarantee a minimum. Players who want better odds at a rarity without a guaranteed floor use these.

Pack Name	Boosted Rarity	Approx. Boosted Drop Rate (free slots)
Rare Boosted Pack	Rare ⚡	Rare: ~55%, Common: ~38%, Epic: 6%, others unchanged
Epic Boosted Pack	Epic 💥	Epic: ~20%, Rare: ~40%, Common: ~33%, Special: 6%, others unchanged
Special Boosted Pack	Special ★	Special: ~8%, Epic: 10%, Rare: ~35%, Common: ~40%, Legendary: 6%

7.3.3 Lucky Packs

Guaranteed minimum rarity on all slots. Distinctly different from Rarity-Boosted: Lucky replaces the floor, Boosted shifts the table.

Pack Name	Minimum Rarity	Roll above minimum
Lucky Epic Pack	Epic 💥	~80% Epic, 15% Special, 4.5% Legendary, 0.5% God
Lucky Special Pack	Special ★	~85% Special, 13% Legendary, 2% God
Lucky Legendary Pack	Legendary 🌊	~92% Legendary, 8% God

7.4 Pack Upgrade Event (Surprise Pack)

When the player taps to open a Basic Pack, a low-probability check fires before the first card is revealed. If triggered, an 'evolution' animation upgrades the pack visually before the opening sequence begins.

Detail	Value / Note
Trigger moment	On player tap — before first card reveal, after nav bar is hidden
Base probability	~0.5% (to be tuned)
Resulting pack type	Random Special Pack, weighted by tier
Probability by result	Type-Themed: 50% Rarity-Boosted: 30% Lucky Epic: 12% Lucky Special: 6% Lucky Legendary: 2%
Animation	Evolution animation upgrades pack art and border before opening begins

8. Resources

8.1 Coins

Source	Amount
Common card obtained	+1 coin
Rare card obtained	+2 coins
Epic card obtained	+10 coins
Special card obtained	+100 coins
Legendary card obtained	+500 coins
God card obtained	+2,000 coins

8.2 Gems

Source	Amount
Special card obtained	+10 gems
Legendary card obtained	+50 gems
God card obtained	+200 gems
Bonus Pack opened	+5 gems (base, upgradeable)

i Coins and Gems are always awarded regardless of whether the card is new or a duplicate. A future 'Fuse' mechanic will let players trade duplicates for bonus resources.

9. User Level System

Players earn XP by opening packs. Levelling up unlocks the ability to purchase upgrades in the Shop. Levels gate upgrades to prevent early over-optimisation.

9.1 XP Sources

Source	Base XP	Upgradeable?
Basic Pack opened	10 XP	Yes — via Basic Pack XP upgrade
Bonus Pack opened	5 XP	Yes — via Bonus Pack XP upgrade
Special Pack opened	20 XP	No (fixed)

9.2 Level Thresholds (Initial Proposal — subject to tuning)

Level	XP (cumulative)	XP to Next	Unlocks
1	0	100	Starting level. No upgrades available.
2	100	200	Bonus Pack cooldown upgrade (Tier 1)
3	300	400	Basic Pack XP upgrade (Tier 1)
4	700	700	Bonus Pack gem yield upgrade (Tier 1)
5	1,400	1,200	Bonus Pack stored count upgrade (Tier 1)
6	2,600	2,000	Bonus Pack card count upgrade (Tier 1)
7	4,600	3,000	Bonus Pack cooldown upgrade (Tier 2)
8	7,600	4,500	Basic Pack XP upgrade (Tier 2)
9	12,100	6,500	Bonus Pack gem yield upgrade (Tier 2)
10	18,600	9,000	All Tier 3 upgrades unlocked (values TBD)
11+	27,600+	TBD	Further tiers TBD

9.3 Upgrades

Upgrade	Effect	Currency	Level Required
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Bonus Pack Cooldown	Reduces timer (e.g. -30s per tier)	Coins + Gems	Level 2+
Bonus Pack Gem Yield	Adds gems per pack (e.g. +2 per tier)	Coins	Level 4+
Bonus Pack Card Count	Adds cards per pack (e.g. +1 per tier)	Gems	Level 6+
Bonus Pack Max Stored	Adds storable packs (e.g. +1 per tier)	Coins	Level 5+
Bonus Pack Rarity Table	Shifts roll toward better rarities	Gems	Level 7+
Basic Pack XP Reward	Adds XP per pack (e.g. +5 per tier)	Coins	Level 3+

10. UX / UI Design

10.1 Global Style

- Dark base background. Card colours and effects pop against it.
- Card background colour determined by type. Card frame shape determined by rarity.
- Higher rarity = more elaborate frame and more spectacular reveal animation.
- Pack visual identity: collection cover on front face + type-specific borders/forms + opening background per collection and type.
- Variant cards (Special, Legendary, God): progressively more elaborate layouts. God/Legendary cards may be full-art with no traditional border.
- Fonts: bold, modern, with fantasy undertones. Clean but impactful, not heavy serif.

10.2 Global Top Bar

Visible across Pack, Collection, and Shop tabs. Contains (left to right):

- Username — as entered on first launch.
- Level X with XP progress below (e.g. '0/100').
- Coin count with coin icon.
- Gem count with gem icon.

10.3 Screen: First Launch — Username

Shown once, before any other screen. Full-screen prompt asking the player to enter a username. Input is validated (non-empty, max length TBD). Confirms and proceeds to the Main tab. The username can be changed once from Settings > Account.

10.4 Screen: MAIN (Pack Tab)

The primary screen. Shows the Basic Pack ready to open, with contextual background and pack art.

Element	Description
Top bar	Username, level + XP progress, coins, gems.
Background	Full-screen background determined by the active collection and pack type.
Pack visual	The pack occupies the centre of the screen. Front face shows collection cover art. Borders and decorative forms reflect pack type.
Collection selector	Placeholder for future multi-collection toggle (TBD — not in v1).
'Open a Pack' button	Large, prominent button. Triggers pack opening flow.

Bottom nav bar	Hidden from the moment 'Open a Pack' is tapped until the summary screen is dismissed.
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10.5 Screen: CARD REVEAL (during pack opening)

Replaces the Main screen after tapping 'Open a Pack'. Bottom nav bar is hidden throughout.

Element	Description
Top bar	Remains visible. Username, level, coins, gems.
Background	Same collection/type background as Main screen.
Card	Centred. Shows: card name (top), card illustration (centre), description text (below image — Common/Rare/Epic only), card number bottom-left, rarity icon bottom-right.
'New!' badge	Appears on new cards. Shows copy count (e.g. ×3) on duplicates.
Tap interaction	Tap anywhere to proceed to the next card. After the 5th card, proceeds to the Summary screen.
Surprise evolution	If Pack Upgrade Event triggers: evolution animation plays before card 1 is shown. Pack upgrades visually.

i Cards are pre-sorted in ascending rarity order (invisibly) before the first reveal. The player always sees the least rare card first and the most rare card last.

10.6 Screen: SUMMARY (after all cards revealed)

Shown after the 5th card is tapped. Displays all 5 cards of the session simultaneously.

Element	Description
Card grid	All 5 cards shown in a row or grid, smaller scale.
'New!' / copy count badges	Shown on each card as during reveal.
'Save' button	Persists all card results to PlayerState. Awards Coins, Gems, and XP. Returns player to Main screen and restores nav bar.

10.7 Screen: COLLECTION

Scrollable grid of all cards in the active collection. Shows unlocked cards in full colour and locked cards as black silhouettes of their card shape.

Element	Description
Top bar	Username, level, coins, gems.
Collection header	Collection title, and progress counters per rarity: 🌟 X/Y ⚡ X/Y 🌟 X/Y ★ X/Y 💧 X/Y 🏹 X/Y (obtained / total for each rarity).
Filter button	Top-right. Opens filter panel to filter cards by rarity and/or type.
Card grid	4 columns. Cards coloured by type if unlocked; black silhouette if locked.
Unlocked card	Shows card illustration thumbnail, card name, rarity icon.
Locked card	Black silhouette of card frame shape. Card name may be shown (TBD).
Tap on card	Opens card detail view (full card, description, copy count, rarity, type, collection number).

10.7.1 Collection Sort Order

Cards are sorted by the following logic — this order is fixed and not affected by the filter (filter only hides, does not reorder):

Group	Order
1. Commons by type	All Common cards, grouped by type in type order (Type 1 all commons, Type 2 all commons, ...).
2. Rares by type	All Rare cards, grouped by type in type order.
3. Epics by type	All Epic cards, grouped by type in type order.
4. Specials by type	All Special cards: first all Special Type 1, then all Special Type 2, etc.
5. Legendaries by type	All Legendary cards: first all Legendary Type 1, then all Legendary Type 2, etc.
6. Gods by type	All God cards: first God Type 1, then God Type 2, etc.

10.8 Screen: SHOP

Scrollable vertical list of all available upgrades. No separate sections per pack type — all upgrades shown in a single list. Locked upgrades (level requirement not met) are shown greyed out with the required level displayed.

Element	Description
Shop header	'SHOP' title on dark background.
Upgrade row	Each upgrade is a horizontal card row: left side shows upgrade name, icon, and current → next value (e.g. '10 min → 9:30 min'). Right side

	shows the upgrade button with cost (Coins and/or Gems icon + amount).
Locked upgrade	Greyed out row. Upgrade button replaced with 'Level X required' label.
Max tier upgrade	Upgrade button replaced with 'MAX' indicator. No further upgrade possible.
Bonus Pack timer	Bonus Pack ready countdown or 'Ready!' status shown at the top of the shop list.

i The upgrade button must show the exact cost and the precise before → after improvement for the next tier (e.g. 'Cooldown: 10:00 → 9:30', 'Gems per pack: 5 → 7'). Players must be able to make an informed decision without tapping into a sub-screen.

10.9 Screen: SETTINGS

A list of navigation rows. Each row opens a dedicated sub-screen for that specific task. No inline controls on the Settings screen itself.

Row	Icon	Sub-screen content
Language		Select between English and Español. More languages in future.
Google Play		Connect account for cloud save.
Account		Change username (allowed once only). Current username displayed. If already changed: field is read-only with a note.
Terms of Use		Link to external URL or in-app text screen.
Privacy Notice		Link to external URL or in-app text screen.

11. Card Opening — Complete UX Flow

Step	Trigger	Description
1. Main screen	—	Player sees pack with collection cover and type-specific border. 'Open a Pack' button visible. Nav bar visible.
2. Tap 'Open a Pack'	Player tap	Nav bar hides immediately. Pack Upgrade Event check fires.
3a. No event	~99.5% of taps	Standard pack opening animation begins.
3b. Surprise event	~0.5% of taps	Evolution animation upgrades pack visual before opening. Resulting pack type determined by weighted roll.
4. Opening animation	After step 3	Pack-specific animation plays. Cards are pre-sorted invisibly in ascending rarity order.
5. Card 1 reveal	Player tap	Least rare card flips. Description shown (if Common/Rare/Epic). 'New!' or copy count badge appears.
6. Cards 2–4 reveal	Player tap each	Same as above, increasing rarity.
7. Card 5 reveal	Player tap	Most rare card flips. Most spectacular animation for the session.
8. Summary screen	After card 5 tap	All 5 cards visible simultaneously. 'Save' button.
9. Save	Player tap 'Save'	Cards, Coins, Gems, XP applied to PlayerState. Returns to Main screen. Nav bar restores.

12. Future Features (Roadmap)

Feature	Notes
Pack Upgrades	Further tiers beyond Level System v1.
Fuse / Duplicates	Trade duplicates for Coins, Gems, or crafting materials.
Ad Rewards	Optional video ads for packs, gems, or coins.
Picks	Choose 1 card from 5 randomly presented options.
Scratch Cards	Scratch-and-win mechanic for bonus rewards.
More Collections	New card pools, themes, and fully real-world types (e.g. 'Dogs').
Collection Selector	Toggle between available collections on the Main screen.
Daily & Weekly Missions	Short-term engagement goals.
Infinite Achievements	Scaling achievement system with ongoing rewards.
More Tabs	Additional navigation tabs as features grow.
Real-World Type Collections	Fully non-fantastical types (e.g. dog breeds, bird species, landscapes).

i First public release targets inclusion of all roadmap features. Early internal builds may omit them.

Changelog

Version	Date	Summary
0.1	Initial	First idea draft — base concept, stack, initial types and rarities.
0.2	2026-04-12	Full expansion: 10 types, card examples, pack system, SlotRule, Pack Upgrade Event, Bonus Pack, resource table, duplicate policy, visual style, roadmap.
0.3	2026-04-12	Card catalogue with rarities + variant system. User Level system with XP, thresholds, and upgrade unlock gates. Pack rules refined (no duplicates per session, ascending rarity reveal order). Rarity-Boosted vs Lucky Packs clearly separated. All Bonus Pack parameters made upgradeable. Card scope expanded to real animals, plants, objects, landscapes. Pack Upgrade Event probability table defined.
0.4	2026-04-13	Full UX/UI section integrated: First-launch username flow (changeable once), card description field (unique text per Common/Rare/Epic card, absent on Special/Legendary/God), card collection number + rarity icon on card face, pack visual identity model (cover + borders + background), rarity icon system (🌸⚡💥🌟🔮🌀), Collection screen sort order defined, Shop screen upgrade row design with before→after values, Settings sub-screen structure, complete 9-step card opening flow, nav bar hide/restore behaviour. Card descriptions added to full catalogue.